

Classifying the Limits of Asteroid Impact Prevention:
The Threshold Between Stability and Disaster

Rushi Parmar

JGE378 Natural Hazards

Dr. Justin Murfitt

June 17, 2026

1 Introduction

Asteroids are rocky, ancient debris left behind from the formation of our solar system. Asteroids are diverse, coming in all sorts of shapes, sizes, eccentric orbits, velocities, and material compositions. (European Space Agency, 2022) In the view of hazards, the properties of an asteroid are a crucial factor in assessing their risk. Although most of the asteroids reside in the asteroid belt, which is a region between Mars and Jupiter, some asteroids have very eccentric orbits and may approach the orbit of Earth and pose a threat which we must investigate. Objects such as asteroids and comets that reside near Earth or are expected to approach Earth are called near-Earth objects (NEOs). The term impactors refers to asteroids that intersect with Earth's trajectory. (European Space Agency, 2022)

Despite asteroid related disasters being very rare, some can cause globally catastrophic damage. For example, the asteroid impact event that took out the dinosaurs, known as Chicxulub impact, is widely regarded as one of the driving factors of the extinction of the dinosaurs 66 million years ago. The asteroid that caused this impact is predicted to be $\sim 10\text{-}15\text{km}$ in diameter, and left behind a 200km wide crater, with debris flying as fast as 11km/s . As a result, the Earth was rapidly blanketed by a thick cloud of soot, dust and sulfate aerosols. The severe environmental changes due to this dust cloud is what caused the mass extinction. (Morgan et al., 2022)

Due to the extinction level threat that asteroids pose, it is crucial for humanity to be prepared for the worst. The main focus of this paper is not only to study asteroid threats, but to further analyze our current technological and theoretical methods through the lens of asteroid impact mitigation. That is, this paper defines a class of preventable scenarios by identifying the top variables involved in asteroid mitigation and uses that definition to evaluate our present limits in planetary defense. The complex system of mitigation will be simplified into a three step pipeline: detection, risk assessment, and mitigation. For each phase, representative baselines will be chosen to reflect current best-available methods for further simplification. As a result, the final classification will be a conservative estimate of our current technology. Table 1 contains the baselines that will be used, paired with a brief justification. In each phase, we will discuss the chosen baseline, how it is used, and its limitations.

Phase	Baseline	Justification
Detection	NSF–DOE Vera C. Rubin Observatory ¹	Top-of-the-line surveying technology expected to begin full operation in 2026. Features the largest digital camera ever made, preliminary data already showcased new NEO detections.
Risk assessment	NASA JPL CNEOS Sentry	NASA’s primary automated system currently used for continuous impact risk monitoring.
Mitigation	NASA’s DART mission	This mission is the only scientific experiment conducted for asteroid mitigation.

Table 1: Baseline technologies and their justification for each phase of asteroid mitigation.

2 Detection

There are hundreds of ground-based observatories on Earth continuously surveying the sky every night to detect NEOs, which are cataloged into extensive databases. In particular, we will study the soon upcoming Vera Rubin observatory, to understand how a ground-based observatory detects asteroids, what limitations are involved, and what important variables stand out as crucial for mitigation.

With respect to asteroid diameter, Figure 1 presents an overview of asteroid detection completeness and the threats that are posed. Notice that detection completeness drops dramatically for smaller asteroid diameters. Asteroids with diameters 10-50m, which have the lowest detection completeness, still pose a potential risk for human injury due to the loudness of a sonic boom and broken glass, and these asteroids can cause noticeable regional damages. Despite larger asteroids with diameters 1-10km potentially causing national to global devastation, the smaller asteroids may represent a much greater overall risk in the foreseeable future for humans and environments, due to the high count and much higher cumulative probability of impact for smaller asteroids.

¹The primary study used for this phase is written before the Vera Rubin observatory becomes operational, and utilized simulations. However, the observatory is set to begin full operation sometime in 2026, and there is promising preliminary data available.

The Hazard by the Numbers

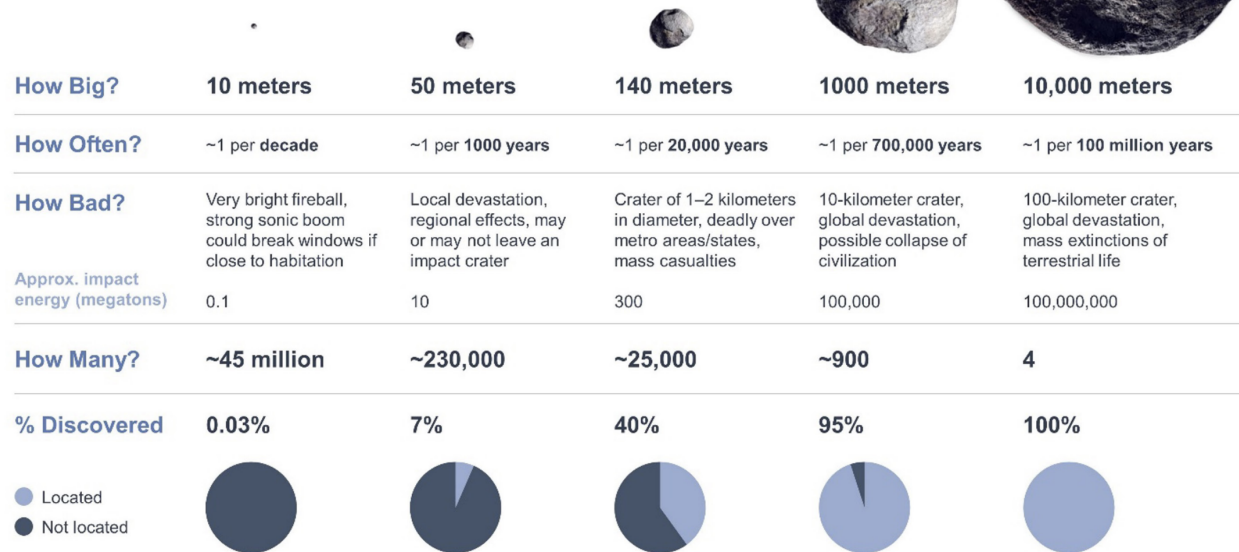


Figure 1: Asteroid hazards and detection completeness with respect to asteroid diameter (National Aeronautics and Space Administration, 2023)

The Vera Rubin observatory is an observatory that is currently going through its final stages of development, and is expected to be one of the most advanced ground-based survey instruments, designed for wide-field imaging of the night sky. (Thomas et al., 2020) In summary, an observatory such as the Vera Rubin Observatory detects asteroids by detecting the faint sunlight reflected off them and identifying their motion across successive images of the sky. (Golovich et al., 2025) Cheng et al. (2026) studies the Vera Rubin observatory's ability to discover imminent impactors, by running survey simulations of 17,000+ potential asteroid impactors, ranging in various properties such as size. However, some limitations on the use of simulation include limited orbital diversity which limits the study of the observatory's response to rarer orbits, and also not considering the sharing of data between different observatories to better map NEOs. (Cheng et al., 2026)

Warning time is a measure of the amount of time left after detection, before an impacting asteroid strikes Earth. Impactors with warning times less than 1 year are close to impossible to prevent. (Tonry, 2010) As shown in Figure 2, the warning times are positively correlated with asteroid size. Notice how 39% of impactors with sizes > 140m are detected over a year in advance, whereas only a small percentage of impactors with sizes < 50m are detected over a year in advance

due to limits in photometric sensitivity. Therefore, asteroid diameter is a critical factor as it is positively correlated with both warning time and detectability. Specifically, an impactor of size $> 140\text{m}$ is favorable for our current methods due to the long warning time and higher probability of detection.

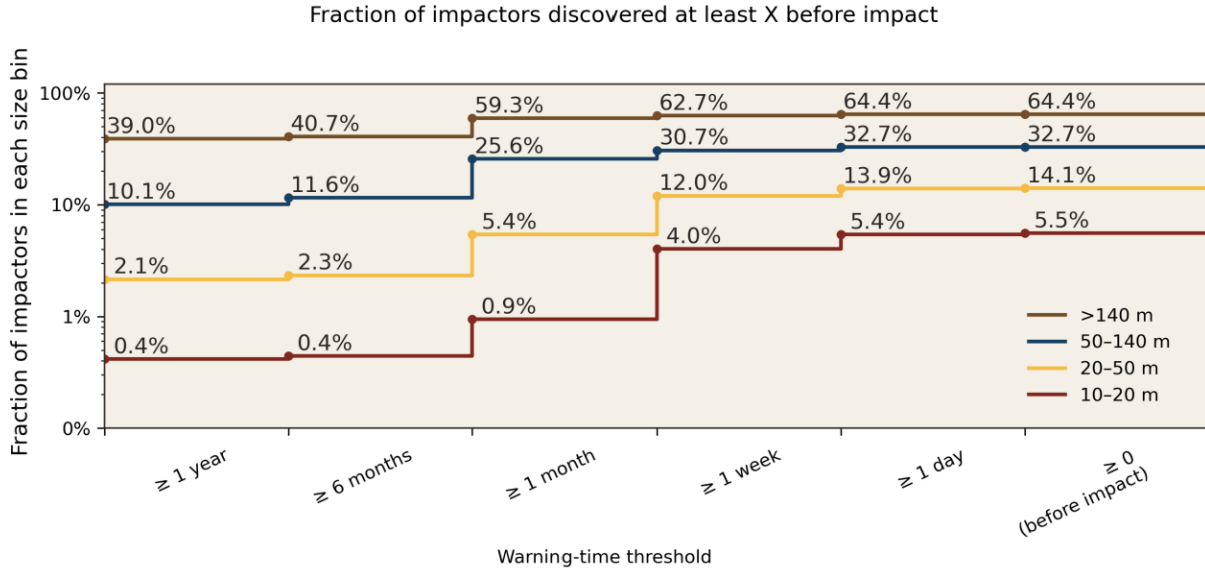


Figure 2: Fraction of impactors, split by size ranges, discovered with respect to warning time (Cheng et al., 2026)

Furthermore, the solar elongation of impactors is another variable to consider in detection. Solar elongation refers to the angular separation between an object and the Sun as seen from Earth. Objects with low solar elongation ($0\text{--}90^\circ$) appear close to the Sun in the sky, whereas objects with large solar elongations ($90\text{--}180^\circ$) appear farther from the Sun. As seen in Figure 3, the frequency of detections increases dramatically for large solar elongations. Majority of the detections occur at solar elongations of $140\text{--}170^\circ$. This makes sense as observatories do not operate during the day because the Sun obstructs the view of the observatory, therefore it is very unlikely to detect an impactor with a small solar elongation. This introduces another technological limitation for ground-based observatories. Therefore, an impactor with solar elongation of $140\text{--}180^\circ$ is favorable.

From this phase, we conclude that *asteroid size*, which is correlated with *warning time*, and *solar elongation* are highly crucial factors. Standard ground-based detection technology is limited by light

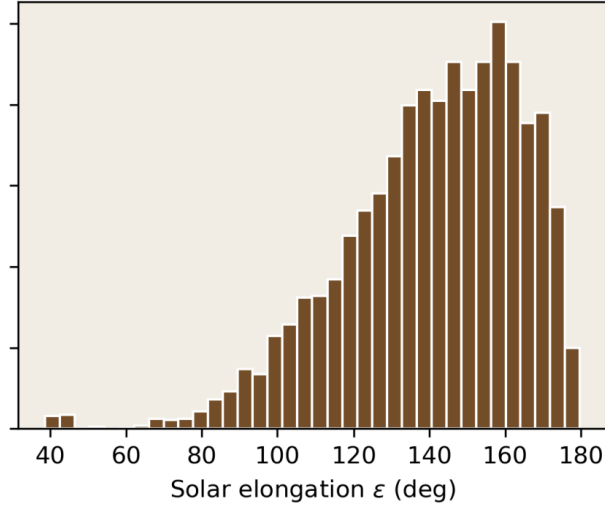


Figure 3: Frequency of impactors detected with respect to solar elongation angle ϵ . The y-axis represents the frequency, although the paper does not explicitly state the numerical scale.

(Cheng et al., 2026)

sensitivity, weather conditions, and the time of day. These limitations can be less severe by cross referencing data from other observatories, and through launching more orbit based observatories. Orbit-based observatories overcome some of these limitations by providing continuous, unobstructed surveying and can detect asteroids with low solar elongations, or asteroids too faint to be seen from Earth. (Lay et al., 2024)

3 Risk assessment

After the detection of an asteroid, it must be studied with further observations to determine its characteristics to a higher accuracy. In particular, its orbit must be modelled accurately to determine a realistic impact probability. In this phase, we will briefly study the baseline technology which is NASA’s Sentry impact monitoring system.

The goal of the Sentry system is to computationally learn the orbit of a NEO and determine an impact probability. Sentry collects the data of NEOs from observatories we discussed earlier, on a daily basis. The orbit of a NEO can never be modelled to full accuracy, therefore some level of uncertainty will always be present. The uncertain orbit forms a probability distribution of possible orbital paths. In other words, there are many acceptable orbits that can be assigned to the asteroid.

In short, the Sentry projects multiple hypothetical orbits in the probability distribution 100 years in advance, and assigns an impact probability based on the fraction of orbits that impact Earth. (NASA, 2026)

The longer a NEO is being tracked for, the more data that becomes available for Sentry, and therefore the less uncertain the NEO's orbit becomes. A recalculation of the orbit can only be done whenever new data becomes available from observatories. (NASA, 2026) Notice that the technological constraints we assessed in the detection phase carry over to risk assessment, meaning the orbits of some asteroids can be learned quicker and easier than others. Uncertainty can be reduced rapidly through frequent, quality observations. A close approach or possible impact of a NEO can be calculated decades in advance if the orbit is well-known. (NASA, 2026)

The impact risks are represented by the Torino scale and the Palermo scale. The Torino scale is meant to communicate risk to the public, which we will discuss because it expresses impact risk in a clear and easily interpretable manner. The scale is based upon impact probability and the consequences of impact. (Binzel, 2000) In Figure 4, we can see that a Torino scale value of above 5 is highly threatening and merit a high priority. A highly uncertain orbit may result in a high Torino scale value, which may seem threatening but further tracking often reduces the risk to a scale value of 0 due to the orbit becoming tightly constrained, making the initial risk an artifact of uncertainty. Therefore, in our classification, a scenario with Torino scale of 5+ that remains consistent despite further tracking is considered beneficial for planetary defense efforts as it represents an identified threat that can be prioritized for mitigation rather than an unexpected impact.

The discussion from this phase further supports the variables and constraints noted from the previous phase, which directly impact how quickly an asteroid orbit can be learned. The uncertainty in the orbit also influences the Torino scale value, and the risk becomes considerable when the Torino scale value is 5 or higher for a well-known orbit.

THE TORINO SCALE <i>Assessing Asteroid and Comet Impact Hazard Predictions in the 21st Century</i>		
Events Having No Likely Consequences	0	The likelihood of a collision is zero, or well below the chance that a random object of the same size will strike the Earth within the next few decades. This designation also applies to any small object that, in the event of a collision, is unlikely to reach the Earth's surface intact.
	1	The chance of collision is extremely unlikely, about the same as a random object of the same size striking the Earth within the next few decades.
Events Meriting Careful Monitoring	2	A somewhat close, but not unusual encounter. Collision is very unlikely.
	3	A close encounter, with 1% or greater chance of a collision capable of causing localized destruction.
Events Meriting Concern	4	A close encounter, with 1% or greater chance of a collision capable of causing regional devastation.
	5	A close encounter, with a significant threat of a collision capable of causing regional devastation.
Threatening Events	6	A close encounter, with a significant threat of a collision capable of causing a global catastrophe.
	7	A close encounter, with an extremely significant threat of a collision capable of causing a global catastrophe.
Certain Collisions	8	A collision capable of causing localized destruction. Such events occur somewhere on Earth between once per 50 years and once per 1000 years.
	9	A collision capable of causing regional devastation. Such events occur between once per 1000 years and once per 100,000 years.
	10	A collision capable of causing a global climatic catastrophe. Such events occur once per 100,000 years, or less often.

Figure 4: An overview of the Torino scale, with the scale number on the left column and the impact scenario on the right column.

(Binzel, 2000)

4 Mitigation

After a high risk impact scenario is determined, actions must be taken to prevent the asteroid impact scenario. Various techniques have been proposed to prevent an asteroid impact which include gravity tractors, nuclear detonation, or kinetic impactors. (Morrison, 2019) Many of these methods remain theoretical and are not scientifically proven. However, the kinetic impactor strategy has been scientifically tested by NASA. NASA's Double-Asteroid Redirection Test (DART), is the only scientific test conducted for the purpose of asteroid mitigation, which will also be the baseline for this phase.

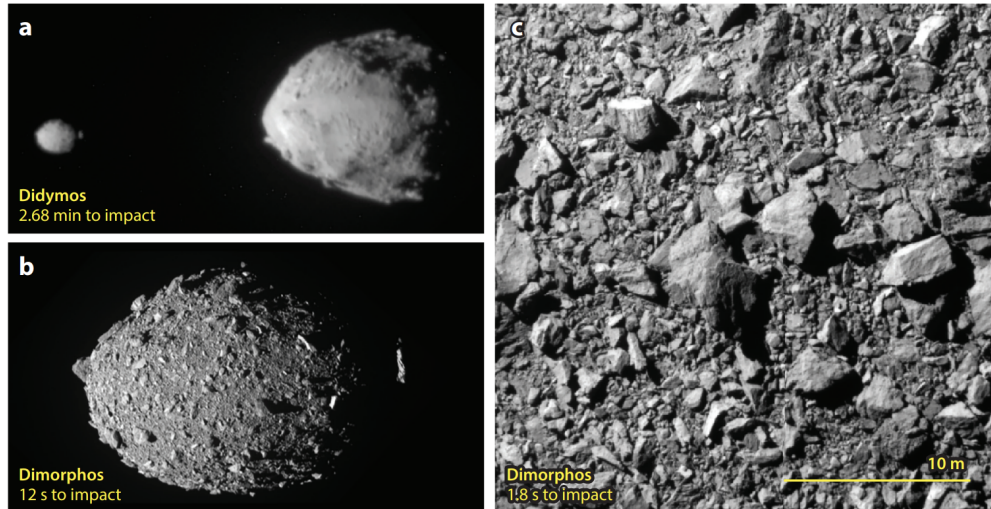


Figure 5: Dimorphos images taken moments before DART impact.
(Rivkin and Chabot, 2026)

In summary, the kinetic impactor strategy involves delivering the momentum of a fast-moving spacecraft onto an impacting asteroid with the goal of perturbing its orbit enough to miss Earth entirely. The DART spacecraft struck its target, an asteroid named Dimorphos, at speeds around 6.1 km/s. The impact reduced the orbital speed of Dimorphos by 2.6 mm/s. The reduction may not sound like much, but overtime the orbit trajectory can deviate hundreds to thousands of kilometers. Dimorphos has a diameter of approximately 150 m, which, as shown in Figure 1, corresponds to a regional-level impact threat. (Rivkin and Chabot, 2026) The DART spacecraft featured a camera and was able to image Dimorphos moments before impact, as seen in Figure 5. Dimorphos is weak, low density, and has a rubble-pile structure as seen in Figure 5.

The results of the test concluded that kinetic impactors are a viable strategy for asteroid mitigation. A warning time of ~ 150 years would be needed for a deflection equal to Earth's diameter, assuming only a single DART impact and a Dimorphos-like impactor. (Rivkin and Chabot, 2026) However, this number can be reduced by utilizing well-timed, frequent series of impacts rather than one. Rivkin and Chabot (2026) proposes that 4 DART-like impacts can reduce the asteroid's velocity by 12 mm/s, giving us a minimum warning time of < 35 years. Furthermore, a scenario requiring a deflection of the entire Earth's diameter is extremely rare, so the warning time required for this mission can be in decades. Assuming an optimistic scenario where the asteroid deviation Δx generally conforms to $\Delta x = \Delta v \cdot t$, deflecting an asteroid by Earth's *radius*

($\Delta x = 6,378$ km) with $\Delta v = 12$ mm/s, would take $t \approx 17$ years. Therefore, we can roughly conclude a warning time of 17 years for a favorable scenario. Of course, this estimate is extremely simplified but this simplification provides a useful first-order estimate of the timescales required for deflection.

Lastly, compositional and structural properties of an asteroid influence the efficiency of a kinetic impactor. Specifically, the momentum transfer depends on asteroid features such as surface strength, internal structure and composition. The weak, rubble-pile like structure of Dimorphos (Figure 5) is beneficial for a kinetic impactor mission. A weaker asteroid releases more ejecta upon impact, resulting in a larger deflection because the outward motion of the ejected material creates a recoil force that accelerates the asteroid in the opposite direction. Solid rock or metallic asteroids will generate less ejecta, making them harder to deflect. Furthermore, a weak asteroid < 80 m in diameter may get broken apart into multiple pieces by a DART impact. (Rivkin and Chabot, 2026) This is not favorable as the debris can still pose a threat.

The mitigation phase introduces a scientifically proven method of asteroid impact mitigation, known as kinetic impactors, by studying the results of NASA’s DART mission. Once again, warning time appears as the leading variable that determines whether or not an impact scenario is preventable. Additional factors such as *asteroid structure*, *asteroid composition* play a role in kinetic impactor efficiency. The size of Dimorphos (~ 150 m) also sits within the favorable asteroid size range we determined in the detection phase from Figure 2.

5 Classification and Discussion

Table 2 presents all the top variables and their favorable values that were determined through the three phases. Recall that a Torino level of 5 or higher is a considerable threat. Given the regional destruction caused by asteroids of sizes 140-150m, a Torino level of 5 or 9 are applicable, but not 8 or 10 since the consequences for level 8 are too little, and the consequences for level 10 are too extreme.

Throughout the three phases, it can be concluded that warning time is the most considerable

factor in asteroid mitigation. Unfortunately, favorable impact scenarios require up to decades of warning time for a feasible mitigation scenario, which heavily restricts the class of asteroids that our current technology can mitigate. The long warning times are primarily a result of our detection limitations. Ground-based observatories cannot survey during the day, and there is a limit to the amount of light they are sensitive to. This result serves as a strong reminder of the need to deploy more orbit based observatories, since the rest of the phases rely on quality, continuous detection. Impactors with diameters 10 – 50m are much harder to defend against because the warning time is far too small for deflection. Despite this class of impactors posing a significant risk for humans, our ability to respond to such impactors is limited, making them one of the most difficult asteroid threats to address. The most likely course of action would be evacuation. (Wheeler et al., 2024)

Variable	Favorable scenario
Warning time	~ 17 years
Torino scale	5 or 9
Asteroid size	140–150 m
Asteroid composition / structure	Weak, low-density, rubble-pile-like structure
Solar elongation	$\sim 140^\circ$ – 180°

Table 2: Summary of variables for a preventable asteroid impact prevention scenario according to the three-phase simplification

6 Conclusion

In conclusion, the study of our current technology through the three phase simplification showcase that warning time is the most critical factor in asteroid impact mitigation. The assessment phase requires adequate warning time to better track and learn the asteroid to accurately determine its impact probability, and the mitigation phase requires adequate warning time to deviate an asteroid away from Earth. The classification indicates a weakness in mitigating smaller sized asteroids, due to technological limitations in ground-based observatories which include observational blind spots and light sensitivity constraints. Orbit-based observatories can handle many of these limitations, by allowing continuous and unobscured viewing of asteroids. Furthermore, the classification represents a very small and rare subset of asteroid impact scenarios, which showcases how unprepared we are for the much larger population of asteroids that fall outside of the preventable class. This paper stresses the need to launch more orbit-based observatories, demonstrates our

vulnerability to smaller asteroids that remain difficult to detect and mitigate, as well as the other asteroid impact scenarios outside of the narrow preventable class.

References

- Binzel, R. P. (2000). The torino impact hazard scale. *Planetary and Space Science*, 48(4), 297–303. [https://doi.org/10.1016/S0032-0633\(00\)00006-4](https://doi.org/10.1016/S0032-0633(00)00006-4)
- Cheng, Q., Scolnic, D., Kurlander, J. A., Chow, I., & Fernandes, M. B. (2026). Assessing the vera rubin observatory’s ability to discover asteroid impactors before they collide with earth. <https://arxiv.org/abs/2601.16255>
- European Space Agency. (2022). Asteroids [Accessed: 2026-05-22]. https://www.esa.int/Science_Exploration/Space_Science/Asteroids
- Golovich, N., Steil, T., Geringer-Sameth, A., Iwabuchi, K., Dozier, R., & Pearce, R. (2025). Survey-wide asteroid discovery with a high-performance computing enabled non-linear digital tracking framework. *Astronomy and Computing*, 53, 100987. <https://doi.org/10.1016/j.ascom.2025.100987>
- Lay, O., Masiero, J., Grav, T., Mainzer, A., Masci, F., & Wright, E. (2024). Asteroid impact hazard warning from the near-earth object surveyor mission. *The Planetary Science Journal*. <https://doi.org/10.3847/PSJ/ad4d9e>
- Morgan, J. V., Bralower, T. J., Brugger, J., & Wünnemann, K. (2022). The chicxulub impact and its environmental consequences. *Nature Reviews Earth Environment*, 3(5), 338–354. <https://doi.org/10.1038/s43017-022-00283-y>
- Morrison, D. (2019). Overview of active planetary defense methods. In *Space and society* (pp. 113–121). Springer. <https://doi.org/10.1007/978-3-030-01000-3>
- NASA. (2026). *Sentry: Earth impact monitoring* [Center for Near-Earth Object Studies (CNEOS)]. Retrieved June 13, 2026, from <https://cneos.jpl.nasa.gov/sentry/intro.html>
- National Aeronautics and Space Administration. (2023). *Nasa planetary defense strategy and action plan: In support of the national preparedness strategy and action plan for near-earth object hazards and planetary defense*. NASA Planetary Defense Coordination Office.

- Rivkin, A. S., & Chabot, N. L. (2026). The dart mission and advancements in planetary defense. *Annu. Rev. Earth Planet. Sci.* 2026, 54. <https://doi.org/10.1146/annurev-earth-032524-125929>
- Thomas, S. J., Barr, J., Callahan, S., Clements, A. W., Daruich, F., Fabrega, J., Ingraham, P., Gressler, W., Munoz, F., Neill, D., Ribeiro, T., Sebag, J., Serrano, E., Stalder, B., Tighe, R., Vucina, T., & Xin, B. (2020). Vera C. Rubin Observatory: telescope and site status. In H. K. Marshall, J. Spyromilio, & T. Usuda (Eds.), *Ground-based and airborne telescopes viii* (p. 114450I, Vol. 11445). SPIE. <https://doi.org/10.1117/12.2561581>
- Tonry, J. L. (2010). An early warning system for asteroid impact. *Publications of the Astronomical Society of the Pacific*, 123(899), 58. <https://doi.org/10.1086/657997>
- Wheeler, L., Dotson, J., Aftosmis, M., Coates, A., Chomette, G., & Mathias, D. (2024). Risk assessment for asteroid impact threat scenarios. *Acta Astronautica*, 216, 468–487. <https://doi.org/10.1016/j.actaastro.2023.12.049>